IVI INFOTAINMENT SYSTEM

Conceptual Architecture of an In-Vehicle Infotainment System Integrating Media, Navigation and Projection

Module 6 – IVI In-Vehicle Infotainment Systems

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1. Introduction

An In-Vehicle Infotainment (IVI) system is a system designed to provide information and entertainment functions inside a vehicle.

This project presents a conceptual architecture of an IVI system integrating three major functions:

- Media
- Navigation
- Projection

The architecture demonstrates how these components communicate with a central IVI processor and how information is delivered to the vehicle display and audio system.

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2. Objective

The main objectives of this project are:

1. To design a conceptual architecture of an IVI system.
2. To integrate media, navigation and projection features.
3. To identify the major components of the IVI system.
4. To show the interfaces between different components.
5. To demonstrate the data flow between the system components.

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3. Proposed IVI System Architecture

The proposed IVI system consists of a central IVI processor connected to different functional components.

The major components are:

- User Interface
- IVI Processor

- Media System
- Navigation System
- Projection System
- Vehicle Data
- Display System
- Audio System
- Speakers

The IVI processor acts as the central component and coordinates the different services.

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4. System Components

4.1 User

The user represents the driver or passenger interacting with the IVI system.

The user can provide input through:

- Touch
- Voice
- Buttons

The user receives information through the display and audio output.

4.2 IVI Processor

The IVI Processor is the central controller of the proposed system.

Its functions include:

- Processing user requests
- Managing different IVI services
- Coordinating data flow
- Connecting media, navigation and projection services
- Sending information to display and audio systems

4.3 Media System

The Media System handles entertainment-related functions.

It can include:

- Music

- Video
- Radio
- USB media
- Bluetooth media

The Media System receives requests from the IVI Processor and provides media information or content.

4.4 Navigation System

The Navigation System provides navigation-related information.

It can use:

- GPS/location information
- Maps
- Route information
- Traffic information

Navigation information can be sent to the display for the user.

4.5 Projection System

The Projection System provides connectivity with an external device such as a smartphone.

It can support:

- Smartphone content
- Applications
- Media
- Screen projection

The projection component communicates with the IVI Processor and sends projected content toward the display system.

4.6 Vehicle Data

Vehicle data provides information related to the vehicle.

Examples include:

- Vehicle status
- Speed
- Gear

- CAN bus data
- Location/GPS information

The vehicle data is communicated with the IVI Processor through a vehicle interface.

4.7 Display System

The Display System presents visual information to the user.

It can display:

- Navigation information
- Maps
- Media information
- Projected content

4.8 Audio System

The Audio System processes audio information received from the IVI system.

It manages:

- Audio processing
- Volume control
- Audio output

The final audio output is delivered through speakers.

4.9 Speakers

Speakers provide the final sound output to the user.

They can be used for:

- Music
- Navigation instructions
- Other audio information

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5. Component Interfaces

The conceptual architecture includes communication interfaces between major components.

Component Connection	Interface

User ↔ IVI Processor	Touch / Voice / Buttons
Vehicle Data ↔ IVI Processor	CAN / Ethernet
Media ↔ IVI Processor	IPC / AIDL
Navigation ↔ IVI Processor	IPC / AIDL
Projection ↔ IVI Processor	IPC / AIDL
IVI Processor ↔ Display	HDMI / LVDS
IVI Processor ↔ Audio System	Audio interface

These interfaces represent the communication paths between the major components of the conceptual architecture.

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6. Data Flow

The general data flow of the IVI system is:

User Input → IVI Processor → Required Service → Display/Audio Output

The main data flow can be divided into the following stages.

Media Flow

User → IVI Processor → Media System → Display/Audio System → User

When the user selects media, the request is processed by the IVI Processor and the required media information is provided through the appropriate output.

Navigation Flow

User → IVI Processor → Navigation System → Display → User

When navigation is requested, the navigation system provides location and route information, which is presented on the display.

Projection Flow

External Device → Projection System → IVI Processor → Display

Content from an external device is communicated through the projection system and presented through the IVI display.

Audio Flow

IVI Processor → Audio System → Speakers

Audio information is processed by the Audio System and delivered through the speakers.

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7. Working Principle

The system works as a centralized infotainment architecture.

First, the user provides an input using touch, voice or buttons.

The input is received by the IVI Processor. The processor identifies the required service and communicates with the corresponding component.

For media requests, the Media System handles the media-related information.

For navigation requests, the Navigation System provides navigation information.

For projection requests, the Projection System communicates with the external device and provides the required content.

The resulting information is then sent to the Display System and/or Audio System.

Finally, the user receives visual information through the display and audio information through the speakers.

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8. Architecture Diagram

The architecture diagram represents the relationship between:

- User
- IVI Processor
- Media System
- Navigation System
- Projection System
- Vehicle Data
- Display
- Audio System
- Speakers

Figure 1: IVI System Architecture Diagram

The architecture diagram is provided separately in the project repository as:

IVI_Architecture_Diagram.png

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9. Data Flow Diagram

The data flow diagram demonstrates the movement of information through the IVI system.

The major flow is:

User Input → IVI Processor → Media / Navigation / Projection → Display / Audio → User

Figure 2: IVI Data Flow Diagram

The data flow diagram is provided separately in the project repository as:

IVI_Data_Flow_Diagram.png

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10. Advantages

The proposed conceptual architecture provides:

1. Integration of media, navigation and projection.
2. Centralized coordination through the IVI Processor.
3. Clear communication paths between components.
4. Both visual and audio output.
5. A modular structure that can be extended with additional services.

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11. Limitations

This project represents a conceptual architecture.

It does not implement a complete physical IVI system or a production automotive infotainment platform.

The architecture is intended to demonstrate the components, interfaces and data flow of an IVI system.

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12. Conclusion

This project presents a conceptual architecture of an In-Vehicle Infotainment system integrating media, navigation and projection features.

The architecture identifies the major components, their interfaces and the flow of information between them.

The central IVI Processor coordinates the different services and provides the required information through the display and audio systems.

Thus, the project demonstrates the basic architecture and interaction of the major components of an IVI system.
